Kazakh-British Technical University

Data collection and preparation

STUDENTS' INDEPENDENT STUDY (SIS) №1

Countries-Population, Life quality and economy

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Almaty

Topic: Countries, population and life quality

1. Intro

The aim of this project is to explore the relationship between population, economic, health and quality of life across different countries. To achieve this, were collected and analyzed real-world data from two sources: World Bank API and World Happiness Report.

2. API and Data cleaning

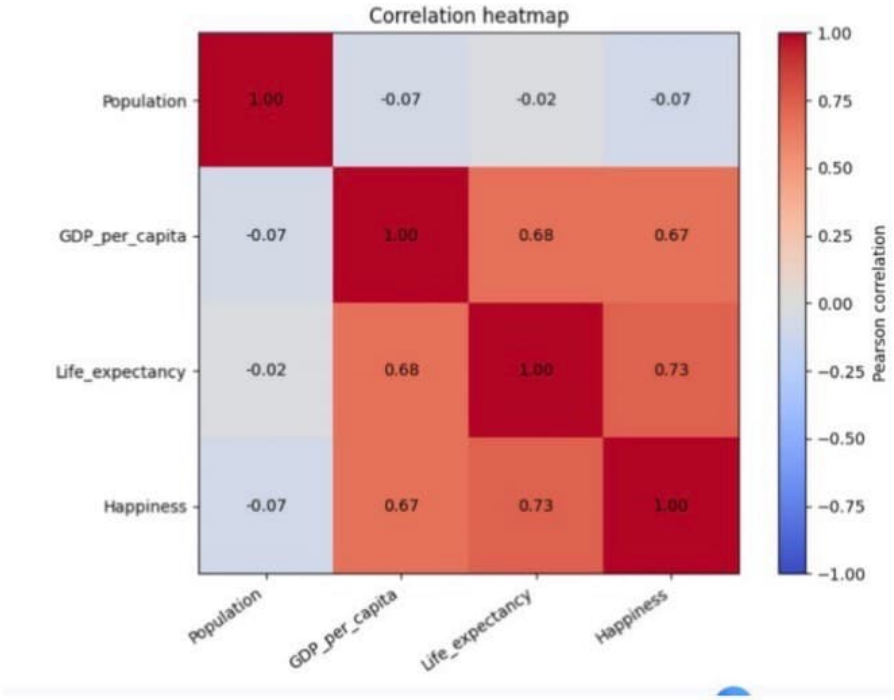
For the first stage of the project data was collected from the World Bank API using Python. Three indicators were selected for 2023: total population GDP per capita and life expectancy at birth. The `requests` library was used to send HTTP GET requests and retrieve the data in JSON format. Each indicator was initially stored in a separate DataFrame, and the three tables were merged using the Country code column as common key with left join. The initial dataset contained 266 records, including countries, territories and aggregates. Regional aggregates were excluded to focus on individual countries and economies, reducing the dataset to 217 records. Data cleaning included checking for missing values, duplicates and data types. Thirteen records with missing GDP per capita values were removed and duplicate country codes were checked. As a result the final dataset contained 204 countries and economies, 5 columns, with no missing values. The cleaned dataset was saved as `world_bank_cleaned_2023.csv` for further integration with the happiness data and exploratory data analysis

3. Web Scraping and Data Cleaning

For the second stage of the project, data was collected from Wikipedia's World Happiness Report 2024 using Python's `requests` and `BeautifulSoup` libraries. Before scraping, the website's robots txt file was checked to ensure that the data collection process followed the website's crawling guidelines. The extracted dataset contained 143 countries with happiness scores ranging from 0 to 10. After extraction the data was cleaned and standardized. Countries were also classified into three happiness levels High, Medium and Low. Country names were adjusted to match World Bank dataset, allowing the two datasets to be merged more easily. According to the extracted data, Finland ranked first with a happiness score of 7.74, while Afghanistan ranked last with a score of 1.72. Nordic countries generally occupied the highest positions in the ranking. Kazakhstan ranked 49th with a happiness score of 6.19.

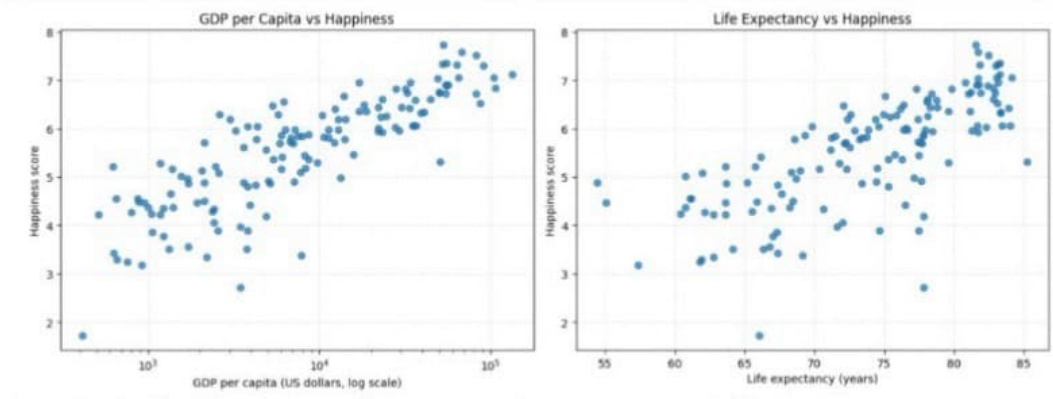
4. Data merging, EDA and visualisations

The cleaned World Bank and World Happiness Report datasets were merged using country names as a common key. The resulting dataset contained **141** countries.



The results showed a strong positive correlation between life expectancy and happiness (r = 0.73) GDP per capita also showed a positive correlation with happiness (r = 0.67) suggesting that countries with higher economic output per person generally reported higher happiness scores

positive correlation was observed between GDP per capita and life expectancy (r = 0.68). Scatter plots, a population bar chart, and a correlation heatmap were used to visualize these relationships



Overall the findings suggest that economic prosperity and life expectancy are more closely associated with happiness than population size.

5. Key insights and conclusion

   Life Expectancy and Happiness: Life expectancy showed the strongest positive correlation with happiness ($r = 0.73$), suggesting that countries with longer life expectancy generally report higher happiness scores.

   GDP per Capita and Happiness: A positive correlation ($r = 0.67$) indicates that economic prosperity is associated with higher levels of happiness.

   Population and Happiness: Population size showed almost no linear correlation with happiness ($r = -0.07$), suggesting that larger populations do not necessarily mean happier societies.

   GDP per Capita and Life Expectancy: A positive correlation ($r = 0.68$) suggests that wealthier economies tend to have longer life expectancy.

   Happiness Rankings: Finland ranked as the happiest country with a score of 7.74, while Afghanistan had the lowest score of 1.72. Kazakhstan ranked 49th with a score of 6.19

The findings suggest that life expectancy and economic prosperity are more strongly associated with happiness than population size. Among the analyzed indicators, life expectancy demonstrated the strongest relationship with happiness